

DAY 9 INTERNSHIP REPORT — DESCRIPTIVE STATISTICS & DATA PROFILING

Prepared by: Prasanna Shetty | Date: 16-09-2026 | Dataset: Iris Dataset (150 entries, 5 features)

Executive Overview

On Day 9 of the internship, the focus was on implementing dynamic Python profiling and calculating descriptive statistics to evaluate feature distributions across numerical columns. Using pandas, an automated classification pipeline was built to segregate continuous measures from target classes, followed by statistical evaluation (central tendency, percentiles, and dispersion metrics) to identify underlying data clusters.

1. Dynamic Column Classification & Profiling

To prevent non-numeric or unique identifier keys from distorting statistical summary outputs, a custom profiling function `classify_columns()` was developed to dynamically assign roles based on data types and unique value counts.

Column Name	Dtype	Unique Values	Assigned Role
sepal length (cm)	float64	35	KEY numeric (measure / amount / count)
sepal width (cm)	float64	23	KEY numeric (measure / amount / count)
petal length (cm)	float64	43	KEY numeric (measure / amount / count)
petal width (cm)	float64	22	KEY numeric (measure / amount / count)
target	object	3	Categorical / text / date

2. Comprehensive Descriptive Statistics Output

Full statistical profiling was calculated across all key numeric variables, incorporating central tendency, dispersion, and percentile distributions.

Feature	Mean	Med	Mode	Std Dev	Min	Q1	Q3	P90	Max
sepal length (cm)	5.84	5.80	5.00	0.83	4.30	5.10	6.40	6.90	7.90
sepal width (cm)	3.05	3.00	3.00	0.43	2.00	2.80	3.30	3.61	4.40
petal length (cm)	3.76	4.35	1.50	1.76	1.00	1.60	5.10	5.80	6.90
petal width (cm)	1.20	1.30	0.20	0.76	0.10	0.30	1.80	2.30	2.50

3. Tabular Analytical Findings & Insights

Feature	Key Metric Value	Analytical Finding & Business Implications
Sepal Length	Mean: 5.84 cm Mode: 5.00 cm	Highlights a specific cluster (mostly Setosa), but is less useful overall because the remaining observations are widely dispersed across the upper range.
Sepal Width	Mean: 3.05 cm Mode: 3.00 cm	Demonstrates a strong, distinct peak. A massive concentration of observations across species converges exactly at 3.00 cm width, explaining the tight standard deviation (0.43 cm).
Petal Length	Mode: 1.40 cm Mean: 3.76 cm Median: 4.35 cm	Crucial Finding: Exposes the multimodal nature of the dataset. 1.40 cm represents the tight Iris-Setosa cluster. The mode completely contradicts the global mean and median, mathematically proving that a distinct subgroup dominates the lower range.
Petal Width	Mode: 0.20 cm Mean: 1.20 cm Median: 1.30 cm	Crucial Finding: Reinforces the Setosa isolation. The most frequent petal width is tiny (0.20 cm), providing concrete evidence that relying solely on global statistical aggregates hides natural species boundaries.